

Math 110 – Homework 9 Answers

Math 110 – Fall 2026

Homework 9

Trace and inner products

Suggested completion date: Friday, October 30

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let P be a projection on a finite-dimensional vector space, so that $P^2 = P$. Prove that

$$\operatorname{tr} P = \operatorname{rank} P.$$

You may choose a basis adapted to the decomposition $V = \operatorname{null} P \oplus \operatorname{range} P$.

Problem 2. Let $T \in \mathcal{L}(\mathbb{C}^5)$ have minimal polynomial

$$m_T(x) = (x - 2)^2(x + 1).$$

- (a) Determine all possible algebraic multiplicities of the eigenvalues 2 and -1 .
- (b) Determine all possible values of $\operatorname{tr} T$.
- (c) For each possible trace, exhibit a Jordan normal form realizing it.

Problem 3.

- (a) Prove that every nilpotent operator on a finite-dimensional complex vector space has trace 0.
- (b) Give a nonzero operator with trace 0 that is not nilpotent.
- (c) Give two 2×2 matrices A, B for which $AB - BA \neq 0$, and verify directly that $\operatorname{tr}(AB - BA) = 0$.

Problem 4. For each of the following formulas on $\mathbb{R}^2$, decide whether it defines an inner product. Justify your answer.

$$\langle x, y \rangle_1 = x_1 y_1 + 2x_2 y_2,$$

$$\langle x, y \rangle_2 = x_1 y_1 - x_2 y_2,$$

$$\langle x, y \rangle_3 = (x_1 + x_2)(y_1 + y_2).$$

Problem 5. On $\mathcal{P}_2(\mathbb{R})$ define

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x) dx.$$

- (a) Prove that this is an inner product.
- (b) Compute the norms of 1, x , and x^2 .
- (c) Determine which pairs among 1, x , x^2 are orthogonal.

Problem 6. Use the Cauchy–Schwarz inequality to prove that for all real numbers a, b, c ,

$$(a + b + c)^2 \leq 3(a^2 + b^2 + c^2).$$

Determine exactly when equality holds.

Problem 7 (optional). Prove the parallelogram identity

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$$

in every real or complex inner-product space.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes
